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STT-MRAM heat sink and magnetic shield structure design for more robust read/write performance

Abstract

An STT-MRAM device incorporating a multiplicity of MTJ junctions is encapsulated so that it dissipates heat produced by repeated read/write processes and is simultaneously shielded from external magnetic fields of neighboring devices. In addition, the encapsulation layers can be structured to reduced top lead stresses that have been shown to affect DR/R and Hc. We provide a device design and its method of fabrication that can simultaneously address all of these problems.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6381094	12/2001	Gill	360/125.75	G11B 5/3903
7262069	12/2006	Chung et al.	N/A	N/A
7829980	12/2009	Malla et al.	N/A	N/A
8125057	12/2011	Bonin et al.	N/A	N/A
8194436	12/2011	Fukami et al.	N/A	N/A
9024399	12/2014	Guo	N/A	N/A
9081669	12/2014	Tadepalli et al.	N/A	N/A
9502466	12/2015	Chuang	N/A	H01L 43/08
9911914	12/2017	Annunziata	N/A	N/A
9985199	12/2017	Briggs	N/A	H01L 43/02
2006/0019431	12/2005	Kasko	257/E27.005	H01L 27/222
2006/0273418	12/2005	Chung et al.	N/A	N/A
2007/0058422	12/2006	Phillips et al.	N/A	N/A
2014/0061827	12/2013	Huang	257/E29.323	H01L 43/08
2015/0091109	12/2014	Allinger et al.	N/A	N/A
2015/0243883	12/2014	Swaminathan	118/723R	H01L 21/0228
2015/0287910	12/2014	Lu	257/421	H01L 43/02
2016/0190432	12/2015	Shum	257/427	H01L 27/222
2016/0254209	12/2015	Oohiraki et al.	N/A	N/A
2016/0351792	12/2015	Jiang	N/A	N/A
2017/0186943	12/2016	Annunziata	N/A	H01L 43/08
2018/0182809	12/2017	Liu	N/A	H01L 43/12
2018/0358545	12/2017	Sundar	N/A	C23C 16/505
2019/0207083	12/2018	Zhong et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102016116301	12/2016	DE	N/A

102016117034	12/2016	DE	N/A
20160118386	12/2015	KR	N/A

OTHER PUBLICATIONS

German Patent and Trademark Office Application No. 112018005816.6, German Patent and Trademark Office, Examination search report, dated Jan. 31, 2024, 8 pages. cited by applicant
Chatterjee, Subho et al., "Impact of Self-Heating on Reliability of a Spin-Torque-Transfer RAM Cell", IEEE Transactions on Electron Devices, vol. 59, No. 3, Mar. 2012, pp. 791-799. cited by applicant

Wang, You e al., "Compact Model of Dielectric Breakdown in Spin-Transfer Torque Magnetic Tunnel Junction", IEEE Transactions on Electron Devices, vol. 63, No. 4, Apr. 2016, pp. 1762-1767. cited by applicant

Guo, W. et al., "SPICE modelling of magnetic tunnel junctions written by spin-transfer torque", HAL archives, HAL Id: hal-00569612, Feb. 25, 2011, 16 pgs. cited by applicant

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Background/Summary

PRIORITY DATA (1) The present application is a continuation application of U.S. patent application Ser. No. 17/107,409, filed Nov. 30, 2020, which is a divisional application of U.S. patent application Ser. No. 15/857,782, filed Dec. 29, 2017, each of which is hereby incorporated by reference in its entirety.

BACKGROUND

1. Technical Field

(1) This disclosure relates generally to magnetic storage devices, specifically to STT-MRAM (Spin Torque Transfer-Magnetic Random Access Memory) devices and methods to improve their thermal stability.

2. Description of the Related Art

(2) STT-MRAM is becoming an increasingly promising candidate for future generation non-volatile working memory to replace embedded flash memory and embedded SRAM (Static Random Access Memory). However, there are challenges in scaling down this technology to 20 nm (nanometers) dimensions and beyond. One such challenge is to enhance the thermal stability of smaller MTJ (magnetic tunneling junction) devices, which is one type of storage cell employed in MRAM. Studies have reported that self-heating of the MTJ junction occurs during read/write cycles (see, for example, S. Chatterjee, S. Salahuddin, S. Kumar, and S. Mukhopadhyay, IEEE Transactions on Electron Device, Vol. 59. No. 3, March 2012; Y. Wang, H. Cal, L. Naviner, Y. Zhang, X. Zhao, E. Deng, J. Klein, and W. Zhao, IEEE Transaction on Electron Device, Vol. 63, No. 4. April 2016; W. Guo, G. Prenat, V. Javerliac, M. Baraji, N. Mestier, C. Baraduc, B. Dieny, Journal of Physics D: Applied Physics, IOP, 2010, 43(21), pp. 215001.)

(3) Self-heating is expected to become even more of a problem as both read/write speed and pattern density increase. On the one hand, self-heating can help reduce the switching current but on the other hand it can also reduce the device thermal stability and even the device reliability. Another challenge for STT-MRAM is the switching disturbance caused by stray magnetic fields from neighboring devices. These and other problems, such as undesirable stresses, related to STT-MRAM operation have been considered in the prior art, for example, in all of the following: U.S.

Patent: 20150091109 (Allinger et al.) U.S. Pat. No. 9,024,399 (Guo) U.S. Pat. No. 7,262,069 (Chung et al.) U.S. Patent Application 2007/0058422 (Phillips et al.) U.S. Pat. No. 8,194,436 (Fukami et al.) U.S. Pat. No. 9,081,669 (Tadepalli et al.) U.S. Pat. No. 8,125,057 (Bonin et al.) U.S. Pat. No. 7,829,980 (Molla et al.) U.S. Patent Application 2006/0273418 (Chung et al.)

(4) It would indeed be desirable to effectively address the problems of self-heating, thermal stability, stresses and switching disturbances. If these problems can be addressed in a combined and efficient manner, it would be even more advantageous. Although the prior arts indicated above have discussed these problems, they have not addressed them in as comprehensive, effective and efficient a manner as in this disclosure.

SUMMARY

(5) A first object of the present disclosure is to provide a method of protecting STT-MRAM devices from adverse thermal effects such as those resulting from self-heating induced by read/write operations.

(6) A second object of the present disclosure is to provide a method to protect STT-MRAM devices from adverse switching effects due to the magnetic fields of neighboring devices.

(7) A third object of the present disclosure is to provide a mechanism for reduction of stress within certain areas of the STT-MRAM device that is a result of the adverse thermal effects.

(8) A fourth object of this disclosure is to use the same heat sink design to serve as a stress buffer for an MTJ device.

(9) A fifth object of the present disclosure is to provide such a method that is able to simultaneously produce all of the above objects.

(10) These objects will be achieved through the design and fabrication of a heat sink structure for STT-MRAM devices that will improve STT-MRAM device thermal stability. This heat sink structure will simultaneously serve as a magnetic shield and stress buffer for magnetic devices. An internal study has shown that top lead stress can affect DR/R and Hc. FIG. 1 shows these results. We are therefore provided with empirical evidence that the same heat sink design can also serve as a stress buffer for an MTJ device, where the stress includes intrinsic film tensile and compressive stresses plus the stress induced by differential expansion/contraction between the BIT line and the overall stack.

(11) The present disclosure provides a design for a heat sink structure and its method of fabrication for an MTJ device, such as an MTJ device that can be integrated into a STT-MRAM device, so that the heat generated during the read/write cycle of such an MTJ device can be dissipated away much more quickly than occurs in an MTJ device that is fabricated using current methods. As a result, the MTJ device so designed and fabricated has its read/write reliability improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1a-1b display data that indicate how top lead stress can affect both DR/R and Hc.

(2) FIG. 2 is a table (Table 1) listing the layer names and their functions corresponding to the “old” MTJ fabrication scheme illustrated in FIG. 4a-4e and currently in use.

(3) FIG. 3 is a table (Table 2) listing the process flow steps corresponding to the fabrication scheme illustrated in FIG. 4a-4e.

(4) FIG. 4a-4e is a schematic illustration showing the present magnetic tunneling junction (MTJ) structure and the fabrication process utilized to fabricate it.

(5) FIG. 5 is a table (Table 3) listing the layer names and their functions corresponding to the presently disclosed fabrication scheme illustrated in FIG. 7 a-7f.

(6) FIG. 6 is a table (Table 4) listing the process flow steps corresponding to the presently disclosed fabrication scheme illustrated in FIG. 7a-7f.

(7) FIG. 7a-7f is a set of schematic illustrations showing the presently disclosed magnetic tunneling junction structure and the fabrication process utilized to fabricate it.

(8) FIG. 8a-8d are a set of schematic illustrations showing an alternative magnetic tunneling junction (MTJ) structure providing equivalent properties to that shown in FIG. 7a-7f and the fabrication process utilized to fabricate it.

DETAILED DESCRIPTION

(9) FIG. 2 illustrates a current integration scheme (i.e., prior art) used to fabricate an MTJ junction for an integrated MTJ device comprising a multiplicity of individually formed and encapsulated junctions. Such devices can be used to form STT-MRAM logic devices. The layer names and general process integration steps are listed in Table 1 (FIG. 2) and Table 2 (FIG. 3) respectively. In the current method, the first step is to deposit the basic unpatterned MTJ film stack and the etch-stop hard masks deposition within them that are used to pattern the stack into a multiplicity of smaller stacks so they can be integrated into a larger device.

(10) Referring to the table shown in FIG. 2 and the corresponding schematic illustration of FIG. 4a, we see that the film stack deposition is a series of 5 layers, 14-10, where 10 is a pin layer, 11 is a barrier layer, 12 is a free layer, 13 is a first hard mask layer, such as Ta, TiN or other conductive material and 14 is a second hard mask layer such as the non-conductive material SiON, or the conductive material TiN. Layers 13 and 14 can be deposited using the same tool that deposits 10-12, or it can be deposited using a different tool. Note that the stack is assumed to be formed on an appropriate substrate such as a conducting bottom electrode (BE) so that when all the processing steps are completed the fabricated device can be easily integrated into desired circuitry.

(11) According to the process steps listed in the table of FIG. 3 and the corresponding illustration in FIG. 4b, step 2 is the deposition of a photo-resist pattern layer, 15, on layer 14. Step 3 is to etch the MTJ deposition, leading to the two separate stacks shown in FIG. 4c. Note, in this description and those to follow we show an initial MTJ stack patterned into two separate stacks. For clarity, the two stacks are shown as isolated, it is assumed, of course, that they are resting on some substrate, but the substrate is not shown in the figure. The example of two stacks is arbitrary and chosen for simplicity and any number of stacks can be processed using these methods.

(12) Step 4 is the deposition of an encapsulation layer (16 in FIG. 4d) which is a layer of dielectric material such as SiN, SiO₂, Al₂O₃, MgO, or the like, deposited to a thickness of between approximately 20-200 Å to protect the patterned MTJ stacks. The encapsulation layer also covers the substrate on which the stacks rest and which is not shown in FIG. 7c. This encapsulation layer, 16, can be deposited in-situ in the same tool that is used to etch the MTJ device or it can be deposited using a separate tool. This encapsulation layer is normally a dielectric material such as SiN, SiO₂, Al₂O₃, MgO, or the like. This encapsulation layer can also be deposited initially as a metal layer and then be oxidized into a dielectric layer. The functionality of this encapsulation layer is not only to insulate the MTJ device from shorting, but also to preserve the magnetic properties and thermal stability of each MTJ device. Therefore, the selection of materials for this layer cannot be some random choice among dielectric materials.

(13) Referring finally to FIG. 4e, there is shown step 5, which is to fabricate the connection of a BIT line, shown as 18, to the MTJ device. This is normally done by first depositing a space-filling interlayer dielectric film (ILD), shown as 17, and then doing chemical mechanical polishing (CMP) to planarize and open the MTJ device. Finally the BIT line 18 is formed to connect to the MTJ. Note that the CMP process removes the upper surface of the encapsulation layer 16 as well as second hard mask, 14, which opens the device to enable an electrical contact between the metal BIT line and first hard mask, 13. Note that the BIT line is normally formed by a dual Cu-damascene process that is well known in the art and will not be described herein.

(14) The encapsulation layer, 16, normally has very low thermal conductivity. The interlayer dielectric material 17 also has very low thermal conductivity. The candidates for layer 17 are often SiN and SiO₂. Due to the low thermal conductivities of the encapsulation layer and the ILD layer,

the majority of the heat generated during read/write processes of the completed devices can only be dissipated by passing through the interface between MTJ and BIT line, **130**, or the interface between the MTJ and BE (bottom electrode), **140**. As the MTJ size decreases, the interface area between MTJ and BIT line and between the MTJ and BE also decreases. As a result, these interfaces become less efficient in dissipating heat, which can become an even worse problem as the read/write speed increases.

(15) FIG. **7a-7f** illustrate the new integration scheme for fabricating a MTJ device that will meet the objects described herein. The table in FIG. **5** and the table in FIG. **6** list the layer descriptions and process steps respectively for ease of discussion. The key differences between the currently used (prior art) method just described and illustrated in FIG. **4a-4e** and the new method about to be described, are step **4** and step **5** of Table **4**, and their corresponding illustrations in FIG. **7d-7e**. In Step **4** shown in FIG. **7d**, instead of depositing the single layer of encapsulation dielectric, **16**, as in FIG. **4d** of the prior art method, two additional encapsulation layers, **19** and **20** are added to augment layer **16**, which is still deposited to a thickness between approximately 20-200 Å. The second encapsulation layer (**19** in FIG. **7d**) is typically a layer of metallic material (electrically conductive or non-conductive and possibly magnetic) with high thermal conductivity, deposited to a thickness of between approximately 20-100 Å, which will act as a heat sink layer. The third encapsulation layer, **20** in FIG. **7d**, is a hard mask layer formed of SiO₂ or SiN to a thickness of between approximately 50-300 Å which is to be used for patterning layer **19**. The process for patterning layer **19** (Step **5** shown in FIG. **7e**) is normally done by a commonly used self-alignment spacer etching method using layer **20** as a hard mask to align the etch, which is guided so that it stops at layer **16** and leaves pieces of layers **20** and **19** along the sidewalls. Note that the etch is a RIE etch of good selectivity between layers **19/20** and layer **16**. An alternative etch scheme that uses different gasses can be used after the etch of layer **20**, in which case layers **20** and **16** can be formed of the same dielectric materials. After the etch, layer **19** will be isolated from the MTJ etch device. Each individual layer **19**, when patterned, will act like a small “bell jar” to surround each MTJ device. Layer **19** will subsequently act as a heat sink layer. After layer **19** layer is patterned in step **5** (FIG. **7e**), an ILD layer **17** will be deposited (step **6** of FIG. **7f**) and followed with a CMP process to open both the heat sink layer, **19**, and the MTJ device **13** at the same time. After that, a similar process like step **5** in FIG. **4e** (prior art method) will be used to fabricate a BIT line (**18** of FIG. **7f**) to electrically connect to the MTJ and layer **19**. Note that the etch process has removed layers **19** and **20** from all but the sides of the two patterned MTJ stacks, while the CMP process removes the tops of layer **16** and all of layer **14** so the BIT line connection can be made.

(16) When magnetic permeable material (such as NiFe, etc.) is used for layer **19**, this layer can then be used as magnetic shield to absorb the stray magnetic flux from adjacent devices and protect the MTJ device. Depending on the magnetic material selection (it should have good thermal conductivity), this structure can serve both as a heat sink and as a magnetic shield. At the same time, the surrounding stress on the MTJ device can be modulated by inserting a layer **19** that is formed of materials of different elastic constants.

(17) Referring next to schematic FIG. **8a-8d**, there is illustrated and described (using the steps in the table of FIG. **6**) an alternative design (second embodiment) of a heat sink and magnetically shielded MTJ device that also meets the described objects set forth above.

(18) FIG. **8a-8d** schematically illustrates this second embodiment of the method which begins where step **3**, shown in FIG. **7c**, ends. We assume that steps **1**, **2** and **3** of this second embodiment are identical to those three steps previously illustrated in FIG. **7a-7c** and described in table of FIG. **6**. FIG. **8a** now immediately follows the structure shown in FIG. **7c** and shows an encapsulation process that replaces that in FIG. **7d** and which was described as step **4** in the table of FIG. **6**. In FIG. **8a** of this alternate embodiment, the patterned MTJ stack of FIG. **7c** has already been encapsulated by layer **16**, which is now followed by successive encapsulations of three additional layers, **19**, **19a** and **20**. Layer **19** is the heat sink layer and layer **19a** is the magnetic shield. These

are now two different layers (**19** and **19a**) which are both formed to thickness between 20-100 Å, whereas previously a single layer could serve as both the heat sink and magnetic shield layer if it was both magnetic and heat conducting. The deposition sequence order between **19** and **19a** can be changed. But in whichever order they are deposited, these two layers are now patterned separately by first patterning layer **19** using a self-aligned etch process, then depositing layer **19a** and etching it away using another photo-etch process.

(19) FIG. **8d** shows an alternative approach to replace FIG. **8c**. The difference here is that the magnetic shield layer is patterned in separate step that removes encapsulation layer **19** so that magnetic shield layer **19a** now remains exposed on top of the MTJ stack where it can contact the BIT line **18**.

(20) As is finally understood by a person skilled in the art, the detailed description given above is illustrative of the present disclosure rather than limiting of the present disclosure. Revisions and modifications may be made to methods, materials, structures and dimensions employed in forming and providing a thermally and magnetically shielded MTJ device, while still forming and providing such a structure in accord with the spirit and scope of the present invention as defined by the appended claims.

Claims

1. A device comprising: a first magnetic tunneling junction (MTJ) stack; a first encapsulation layer disposed on a sidewall surface of the first MTJ stack; a first magnetic shield layer disposed on the first encapsulation layer along the sidewall surface of the first MTJ stack; a second encapsulation layer disposed on the first encapsulation layer along the sidewall surface of the first MTJ stack; a dielectric layer disposed directly on the first encapsulation layer, the magnetic shield layer and the second encapsulation layer such that the dielectric layer physically contacts the first encapsulation layer, the magnetic shield layer and the second encapsulation layer; and a conductive line disposed over the dielectric layer and electrically coupled to the first MTJ stack.
2. The device of claim 1, further comprising a second MTJ stack, wherein the first encapsulation layer extends continuously from the first MTJ stack to the second MTJ stack.
3. The device of claim 1, wherein the first encapsulation layer physically contacts the conductive line.
4. The device of claim 3, wherein the first magnetic shield layer, the second encapsulation layer and the dielectric layer physically contact the conductive line.
5. The device of claim 1, wherein the first magnetic shield layer physically contacts the conductive line, the second encapsulation layer, the first encapsulation layer and the first MTJ stack.
6. The device of claim 5, wherein the magnetic shield layer is positioned between the conductive line and the second encapsulation layer, the first encapsulation layer and the first MTJ stack thereby preventing the second encapsulation layer, the first encapsulation layer and the first MTJ stack from physically contacting the conductive line.
7. The device of claim 1, wherein the first encapsulation layer includes a dielectric material, wherein the first magnetic shield layer includes a first metal material, and wherein the second encapsulation layer includes a second metal material that is different from the first metal material.
8. The device of claim 1, wherein the second encapsulation layer is a first heat shield layer formed of a different material than the first magnetic shield layer.
9. A device comprising: a first magnetic tunneling junction (MTJ) stack; a first encapsulation layer disposed on the first MTJ stack; a first heat shield layer disposed on the first encapsulation layer; a first magnetic shield layer disposed on the first encapsulation layer; a conductive feature disposed directly on the first heat shield layer, the first magnetic shield layer and the first MTJ stack; and an interlayer dielectric fill layer extending continuously from a bottom surface of the conductive feature to a top surface of the first encapsulation layer, the top surface of the first encapsulation

layer facing the bottom surface of the conductive feature.

10. The device of claim 9, wherein the conductive feature is further disposed directly on the first encapsulation layer.

11. The device of claim 9, wherein the first heat shield layer is disposed directly on the first encapsulation layer, and wherein the first magnetic shield layer is disposed directly on the first heat shield layer.

12. The device of claim 9, wherein the first heat shield layer is disposed directly on the first magnetic shield layer, and wherein the first magnetic shield layer is disposed directly on the first encapsulation layer.

13. The device of claim 9, further comprising: a second MTJ stack spaced apart from the first MTJ stack; the first encapsulation extending continuously from the first MTJ stack to the second MTJ stack; a second heat shield layer disposed on the first encapsulation layer that is disposed on the second MTJ stack; a second magnetic shield layer disposed on the first encapsulation layer that is disposed on the second MTJ stack; the conductive feature extending continuously from the first MTJ stack to the second MTJ stack.

14. The device of claim 13, wherein the first heat shield layer is discontinuous with respect to the second shield layer, and wherein the first magnetic shield layer is discontinuous with respect to the second magnetic shield layer.

15. The device of claim 14, wherein the interlayer dielectric fill layer interfaces with both the first heat shield layer and the first magnetic shield layer.

16. A device comprising: a first magnetic tunneling junction (MTJ) stack and a second MTJ stack; a first encapsulation layer disposed directly on the first MTJ stack and the second MTJ stack such that the first encapsulation layer extends continuously from the first MTJ stack to the second MTJ stack; a first heat shield layer disposed on the first MTJ stack and a second heat shield layer disposed on the second MTJ stack; a first magnetic shield layer disposed on the first MTJ stack and a second magnetic shield layer disposed on the second MTJ stack; a conductive feature disposed on and interfacing with the first and second heat shield layers, the first and second magnetic shield layers and the first and second MTJ stacks; and a dielectric fill layer extending continuously from a bottom surface of the conductive feature to a top surface of the first encapsulation layer, the top surface of the first encapsulation layer facing the bottom surface of the conductive feature.

17. The device of claim 16, wherein the dielectric fill layer further extends continuously from the first heat shield layer to the second heat shield layer.

18. The device of claim 16, wherein the dielectric fill layer further extends continuously from the first magnetic shield layer to the second magnetic shield layer.

19. The device of claim 16, wherein the first encapsulation layer is formed of a material selected from the group consisting of SiN, SiO.sub.2, Al.sub.2O.sub.3 and MgO, wherein the first heat shield layer is formed of a material selected from the group consisting of Ti, TiN, Cu, Ta, TaN, W, Al and AN, and wherein the first magnetic shield layer is formed of a material selected from the group consisting of NiFe and CoFe.

20. The device of claim 19, wherein the second heat shield layer is formed of the same material as the first heat shield layer, and wherein the second magnetic shield layer is formed of the same material as the first magnetic shield layer.
